

Vidhi Vazirani

Data Scientist · Machine Learning Engineer

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PROFESSIONAL SUMMARY

Data Scientist with 1.5+ years of full-time experience at Infoblox building production ML systems for cybersecurity infrastructure. Specializes in anomaly detection pipelines, LLM fine-tuning for domain-specific classification, and agentic AI systems deployed in Kubernetes-based production environments. Published researcher with 7 papers and 36+ citations (h-index 4) across IEEE, Springer, and IET venues, spanning cybersecurity, healthcare AI, and computer vision. M.S. Computer Science from Northwestern University (4.0 GPA). Experienced across the full ML lifecycle — from dataset curation and model benchmarking to MLflow-tracked deployment and cross-functional stakeholder collaboration.

EDUCATION

Northwestern University *Sept 2023 – Dec 2024*

Master of Science, Computer Science · GPA: 4.00 / 4.00 · Evanston, Illinois

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Generative Methods, Statistical Learning for Data Analysis, Artificial Intelligence, Data Management, Human-Computer Interaction

NMIMS University *July 2018 – May 2023*

B.Tech, Computer Engineering & MBA, Business Intelligence and Analytics · GPA: 3.83 / 4.00 · Mumbai, India

Relevant Coursework: Data Structures, DBMS, Data Mining, Design and Analysis of Algorithms, Image Processing, Software Engineering, Biometrics, NLP, Business Analytics, Programming for Analytics, Big Data Technologies, Introduction to Marketing, B2B Marketing

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, Java, C, C++, SAS

ML & AI Frameworks: PyTorch, TensorFlow, Scikit-Learn, XGBoost, Isolation Forest, SHAP, BERT, Transformers, CNNs, GRU, UNet, SegNet, AlexNet

LLM & Agents: LLM Fine-tuning (LoRA/QLoRA), RLHF, Instruction Tuning, Quantization (4-bit/8-bit), LangChain, OpenAI API, Anthropic API, Prompt Engineering, RAG, Agentic Pipelines, vLLM, RayServe

MLOps & Infrastructure: Databricks, MLflow, Apache Spark, Spark Structured Streaming, Kafka, Delta Lake, Kubernetes, Docker, Harbor Registry

Observability & Monitoring: Prometheus, PromQL, Grafana, Mimir

Data & Visualization: NumPy, Pandas, Keras, OpenCV, NLTK, SpaCy, Matplotlib, Seaborn, Tableau

Big Data: Hadoop, Hive, Spark

Time Series: ARIMA, SARIMA, ARIMAX, MSTL, STL, Holt-Winters, Prophet, Rolling Z-Score, Rolling IQR

Web & Other: HTML, CSS, JavaScript, PHP, Git, GitHub, Figma

PROFESSIONAL EXPERIENCE

Data Scientist · Infoblox · Santa Clara, CA · On-site *Jan 2025 – Present*

Working on applied ML and AI systems across five major workstreams: anomaly detection, LLM fine-tuning for threat intelligence, agentic AI for on-call operations, security log analytics, and internal AI assistant (BloxMate) infrastructure.

1. Infoblox IQ — Anomaly Detection & Forecasting

- Designed and benchmarked a DNS/DHCP anomaly detection pipeline for IQ telemetry (latency, NXDomain counts, server health), evaluating methods on 15.5M records across 133 accounts and 1,881 OPHs
- Built a two-stage pipeline: Stage 1 uses robust z-scores on residuals for initial flagging; Stage 2 uses XGBoost to reduce false positives — cutting false positive rate by 20%+ with only 0.5 ms latency overhead

- Benchmarked five anomaly detection approaches on a 1.73M-record evaluation subset (57 accounts, 200 OPHs, 8,641 data points per series):
 - Rolling Z-Score: flagged ~0.96% of points — high true-positive rate but noisy with seasonal data
 - Rolling IQR: flagged 2.06–3.67% of points — noisiest baseline, poor for production
 - STL decomposition: flagged ~1.50% of points
 - MSTL with auto period detection: flagged only 0.245% — best false-positive behavior for seasonal series
 - AutoARIMA: flagged 0.439%; fixed ARIMA(1,1,1) and ARIMA(2,1,2) flagged ~0.246%
- Developed a 'super' modified z-score variant that reduced anomaly counts to under 50% of the baseline model (at ~2x runtime cost), and a lighter variant that achieved 20% FP reduction at +0.5 ms
- Logged all experiments to MLflow; compared models on offline fit and production-realistic anomaly detection workloads using rate vs. increase PromQL queries segmented by account and service

2. Lookalike Domain Detection — LLM Fine-Tuning

- Built a supervised instruction-style fine-tuning pipeline to classify lookalike domains and return structured fields: target domain, fuzzer type, and classification label
- Evaluated multiple compact models and quantization strategies: Llama 3.2 1B & 3B (instruct and base), Phi-3.5 Mini, Gemma 2, SmolLM — with 4-bit and 8-bit quantization
- Zero-shot accuracy baselines: 0.47–0.548 across configurations; fine-tuned models reached 0.624–0.848 depending on model and setup
- Best configuration: Llama 3.2 3B Instruct (8-bit) — accuracy 0.848, precision 0.839, recall 0.846, F1 0.843
- Scaled training data pipeline from early experiments to ~1M triplets, then ~1.71–2M triplets; identified ~22k hard-negative and adversarial examples for focused fine-tuning
- Explored multimodal and embedding architectures: Canine, Gemma, CLIP-style combinations, Qwen-based teacher-student fine-tuning
- Gemma 4 matched ~84% of frontier-model banding behavior — used as a cost-saving milestone for the associated CTEM-style evaluation workstream
- Contributed to productionization: env-5 deployments, GPU-hosted open-source model serving via vLLM/RayServe, MLflow-based serving experiments, Databricks pipeline migration, schema changes for lookalike tables, and API endpoint integration

3. On-Call Threat Defense Triage Agent

- Designed a two-stage agentic architecture for Threat Defense alert triage: Stage 1 classifies the problem type; Stage 2 selects and runs the appropriate tools, runbooks, and metric queries to generate RCA-oriented answers
- Built agent prompt and routing logic across td.py (traffic runbook and metric query definitions) and prompts.py (TD agent and TD supervisor prompts)
- Fixed critical production failure: agent was not extracting site IDs responsible for traffic spikes — rewrote query flow to extract IP address first, then map to customer/site context; resolved a collapse bug where site ID defaulted to '172' for all queries
- Handled multi-context reasoning across ATC and AGA environments, Grafana-derived metrics, maintenance schedules, content filter errors, token rate limit errors, and expected traffic spikes
- Managed full deployment lifecycle: built dev images via make build-dev-image-all, retagged and pushed to Harbor registry, redeployed in Kubernetes, and debugged live pods (e.g., oncall-agent-td-77d95885bf-7kdvx) via kubectl and k9s
- Live service endpoint: .../api/poseidon-ds-services/v1/oncall-agent-td/query on env-5; tested with real alert payloads including TCP traffic increase scenarios on SP-1
- Iterated on test-case expansion, fallback classification logic, query performance tuning, and Grafana dashboard integration across multiple sprint cycles

4. Security Log Analysis & Feature Adoption Analytics

- Built Spark Structured Streaming + Kafka ingestion pipelines to pull security logs into Delta tables, using asset bundles for job deployment; documented broker/topic/partition model, checkpointing, and schema definition from Kafka JSON
- Performed large-scale EDA on ingested security logs: ~470k unauthorized events, ~48k null app values, ~49k missing session IDs, ~80k infoblox-domain entries, one CSP account contributing ~10k logs, largest body_bytes_sent value of ~4.89 GB

- Profiled request distributions: predominantly POSTs with status 200; identified 202, 204, 206, 400, 403, 404, 408, 499, 500, and 502 edge cases; primary user agent grpc-go/1.67.1
- Built feature space for downstream dashboards covering endpoint popularity, logs per day, account-level spikes, and customer behavior segmentation — integrated into feature-adoption pipeline
- Studied SecEncoder research direction: BPE tokenization, masked language modeling (BERT/DeBERTa-style), and pretraining at scale (30k-token vocab, 100GB raw data, 77B tokens after deduplication from 1TB → 270GB corpus) as a foundation for a log-native encoder

5. BloxMate — Internal AI Assistant & Evaluation Infrastructure

- Implemented two log-aggregation strategies before the researcher agent step — general normalization and LLM-based summarization — to reduce prompt size and token cost
- Built continuity and personalization features: agent references older related responses and aligns with HR/IT user expectations based on v1 focus-group feedback with Brandon, HR, and IT teams
- Built MLflow-based evaluation framework: created test evaluation datasets for insighter traces, wrote a small service for MLflow evaluations, and updated the DS-owned services inventory for monitoring and registration
- Migrated BloxMate's core model from GPT-4.1 to GPT-5.2; ran stakeholder discussions to prioritize testing scope and agentic flow improvements
- Created infrastructure for disciplined agent iteration: prompt compression, conversation-memory experimentation, service ownership registry, and MLflow-backed quality measurement

Artificial Intelligence Intern · Infoblox · Remote, USA Jun 2024 – Aug 2024

- Fine-tuned LLMs (Llama, Gemma, Mistral) on GPT-4o-generated customer service QnA datasets for domain-specific classification and retrieval tasks
- Designed and maintained automated data reporting pipelines using SQL with real-time Grafana dashboards, directly impacting data-driven product decisions

Teaching Intern · Coding Jr. · Remote, India Apr 2023 – Aug 2023

- Led Python and Data Science instructional sessions for 50+ school students; trained instructors in CSS
- Produced 75+ Python and C++ video lectures; achieved 95% positive feedback rating across 10+ schools

Co-Founder & Head of Technology · Being Ambitious · Mumbai, India Apr 2021 – Dec 2022

- Co-founded a content and media platform; led a team of 7 in content creation and operations
- Interviewed 25+ founders, volunteers, and influencers; helped acquire and onboard an international client
- Developed and optimized the website using HTML, CSS, JS, and PHP — drove a 30% increase in web traffic; led brand communication strategy

Brand Strategy & Social Media Marketing Intern · Jio Creative Labs · Mumbai, India May 2022 – Sep 2022

- Developed brand strategy and executed social media marketing campaigns for a leading Indian media and telecom conglomerate
- Collaborated on website content and brand communication deliverables for client accounts

Head of Content / Website Developer · Being Digital · Remote, India Jul 2020 – Sep 2021

- Head of Content (Oct 2020 – Sep 2021): led editorial strategy, content planning, and team coordination for a digital media internship program
- Website Developer (May – Jun 2021): built and maintained web properties; skills in web content writing and front-end development
- Content Creator (Jul – Sep 2020): produced written and digital content across platforms

PUBLICATIONS · H-INDEX: 4 · TOTAL CITATIONS: 36+

Beyond Boundaries: Unifying Classification and Segmentation in Wildfire Detection Systems Cited by 5

S. Singh, V. Vazirani, S. Singhania, V.S. Suroth, V. Soni, A. Biwalkar et al.

Multimedia Tools and Applications, Vol. 84 (20), pp. 22441–22473, Springer · 2025

Presents a two-stage wildfire detection system combining CNN-based image classification and semantic segmentation of burnt areas. A 22,000-image satellite dataset was curated using Google Earth Engine. A custom CNN outperformed AlexNet with 88.19% classification accuracy; UNet surpassed SegNet in segmentation with a Dice score of 0.6869.

Deep Hybrid Approach with Sequential Feature Extraction and Classification for Robust Malware Detection[Cited by 21](#)

S. Singh, D. Krishnan, V. Vazirani, V. Ravi, SA. Alsuhibany

Egyptian Informatics Journal, Vol. 27, 100539, Elsevier · 2024

Proposes a hybrid deep learning architecture combining GRU, CNN, and Random Forest for malware classification using image representations generated from binary malware files. The sequential feature extraction pipeline achieved 99.51% classification accuracy, outperforming individual model baselines.

Classification vs Clustering: Ways for Diabetes Detection[Cited by 5](#)

S. Singh, V. Vazirani

IEEE 7th International Conference for Convergence in Technology (I2CT), pp. 1–8 · 2022

Comparative study of classification and clustering algorithms for diabetes detection in women. XGBoost achieved the highest performance with 98% average accuracy across two benchmark datasets, outperforming SVM, KNN, Decision Trees, and k-Means variants.

Review of Medical Imaging with Machine Learning and Deep Learning-based Approaches for COVID-19

S. Singh, V. Vazirani, D. Krishnan

Smart Health Technologies for the COVID-19 Pandemic: Internet of Medical Things, IET · 2022

Systematic review of ML and DL methods for COVID-19 detection from MRI, CT scans, and X-ray imaging. Surveys state-of-the-art CNN architectures, compares approaches on accuracy metrics, and identifies open research gaps in data availability, model generalizability, and clinical deployment.

Clinical Decision Support Systems and Computational Intelligence for Healthcare Industries[Cited by 5](#)

S. Singh, A. Biwalkar, V. Vazirani

Knowledge Modelling and Big Data Analytics in Healthcare, pp. 37–63, Taylor & Francis · 2021

Reviews 30+ papers published between 2018 and 2021 on computational intelligence approaches for clinical decision support. Covers ML-based diagnosis for Brain Tumor, Diabetes, Heart Disease, and Pneumonia, with analysis of methods, accuracy benchmarks, and identified limitations in existing approaches.

Clinical Decision Support Systems and Computational Intelligence for Healthcare Industries (Volume 2)

S. Singh, A. Biwalkar, V. Vazirani

Knowledge Modelling and Big Data Analytics in Healthcare: Advances, Taylor & Francis · 2021

Extended companion volume covering advances in computational intelligence for healthcare decision support, with additional coverage of emerging techniques and updated benchmarks in the field.

ACADEMIC PROJECTS

Deep Hybrid Approach for Malware Detection · GRU · CNN · Random Forest · PyTorch2023–2024

- Problem: classify malware from binary files without dynamic analysis, using static image representations
- Converted binary malware files into grayscale image representations; applied sequential feature extraction using stacked GRU layers, followed by spatial feature extraction via CNN
- Combined GRU and CNN features and passed to a Random Forest classifier for final classification
- Achieved 99.51% classification accuracy, outperforming standalone GRU, CNN, and Random Forest baselines
- Dataset: publicly available malware image datasets with multiple malware families; evaluation via accuracy, precision, recall, and F1

COVID-19 Medical Named Entity Recognition · SpaCy · BERT · NLP · HuggingFace2023

- Problem: extract and identify medical entities (diseases, drugs, symptoms, procedures) from COVID-19 research literature
- Applied SpaCy NER pipelines for initial entity recognition across a COVID-19 annotated corpus; fine-tuned entity labels for domain-specific medical terminology
- Fine-tuned BERT (bert-base-uncased) for medical question answering, trained to answer COVID-19-related questions grounded in detected medical entities
- Evaluated NER performance using precision, recall, and F1 at entity level; assessed QA quality on held-out COVID-19 research passages

Wildfire Detection System · CNN · AlexNet · UNet · SegNet · Google Earth Engine · TensorFlow2022–2023

- Problem: detect active wildfires and segment burnt areas from satellite imagery at scale
- Curated a 22,000-image dataset (fire / no-fire classes) using Google Earth Engine by querying latitude, longitude, and acquisition dates across fire-prone regions

- Stage 1 — Classification: trained a custom CNN and AlexNet; custom CNN achieved 88.19% accuracy, outperforming AlexNet
- Stage 2 — Segmentation: trained UNet and SegNet on fire-masked images; UNet achieved Dice score of 0.6869, outperforming SegNet
- Contributed to a peer-reviewed publication in Multimedia Tools and Applications, Springer (2025), cited 5 times

Airline Customer Satisfaction Prediction · *PyTorch · Scikit-Learn · Python2023*

- Problem: predict airline customer satisfaction from survey and flight data
- Performed full data pipeline: cleaning, transformation, feature engineering, and exploratory analysis on a large airline satisfaction dataset
- Implemented and compared four model architectures in PyTorch: Linear Regression, Neural Network, Multilayer Perceptron (MLP), and Support Vector Machine (SVM)
- Evaluated on accuracy, confusion matrix, and training time; neural networks achieved the best overall performance
- Analyzed trade-offs between model complexity, training time, and generalization performance

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Board Member · *Society of Women Engineering (SWE), Northwestern University2023 – 2024*

- Led inclusivity and outreach initiatives; managed public relations and content creation for the committee
- Collaborated with like-minded peers to organize professional development, networking, and community events for women in engineering

Advisor & Head of Disciplinary Body · *FinDrome2021 – 2023*

- Led a team of 50+ students to conduct finance-related seminars, social events, industry talks, and panel discussions
- Advised on strategic direction and enforced institutional norms as head of the disciplinary body